



Technical University of Cluj - Napoca
Computer Science Department

Procesarea Imaginilor

(An 3, semesterul 2)

Curs 1: Introducere



Info

Evaluare

Examen scris – 50% din nota (nota minima 5)

- Bonus la nota examen = $nr_prez * 0.1$ (if $nr_prez > nr_cursuri / 2$)

Laborator + proiect – 50% din nota (nota minima 5)

Prezenta la laborator/proiect – obligatorie!

Laborator – evaluare dupa fiecare laborator + teste / colocviu

Proiect – evaluare dupa fiecare faza de proiectare (specifice, analiza, design logic, implementare, testsare si validare).

Documentatii (curs & laborator)

<http://users.utcluj.ro/~tmarita/IPL/IPCurs/IPCurs.htm>

<http://users.utcluj.ro/~tmarita/IPL/IPLab/IPLAB.htm>

Retea Dorobantilor 71-73: \\D110-13\Users\Public\Public Documents\PI

Paginile personale ale lectorilor / indrumatorilor de laborator



Bibliografie curs

- R.C.Gonzales, R.E.Woods, *Digital Image Processing – 2-nd Edition*, Prentice Hall, 2002.
- E. Trucco, A. Verri, *Introductory Techniques for 3-D Computer Vision*, Prentice Hall, 1998.
- W.K. Pratt, *Digital Image Processing: PIKS Inside, 3-rd Edition*, Wiley & Sons 2001.
- G. X.Ritter, J.N. Wilson, *Handbook of computer vision algorithms in image algebra - 2nd ed*, CRC Press, 2001.
- Frank Y. Shih, *Image Processing And Pattern Recognition - Fundamentals and Techniques*, Wiley & Sons, Hoboken, New Jersey, 2010.
- A. Koschan, M. Abidi, *Digital Color Image Processing*, Wiley & Sons, 2008.
- H.C. Lee, *Introduction to Color Image Science*, Cambridge University Press 2005
- D. Forsyth, J. Ponce, *Computer Vision. A Modern Approach*, Prentice Hall, 2002.
- L. G. Shapiro, G. C. Stockman, *Computer Vision*, Prentice Hall, 2000

Bibl. UTCN:

- S.Nedevschi, "Prelucrarea imaginilor si recunoasterea formelor", *Ed. Microinformatica*, 1997.
- S. Nedevschi, R. Dănescu, F. Oniga, T. Marița, *Tehnici de viziune artificială aplicate în conducerea automată a autovehiculelor*, Editura U.T. Press, Cluj-Napoca, 2012.
- S. Nedevschi, T. Marița, R. Dănescu, F. Oniga, R. Brehar, I. Giosan, C. Vicaș, *Procesarea Imaginilor - Îndrumător de laborator*, Editura U.T. Press, Cluj-Napoca, 2013.
- S. Nedevschi, T. Marita, R. Danescu, F. Oniga, R. Brehar, I. Giosan, S. Bota, A. Ciurte, A. Vatavu, „Image Processing - Laboratory Guide”, UTPress Edition, 2016, ISBN 978-606-737-137-6, <http://biblioteca.utcluj.ro/carti-online.html>



Viziunea artificiala

Vizunea artificiala (Computer Vision)?

Viziunea artificiala este un domeniu/disciplina care **infereaza date/informatie din imagini** cu ajutorul **metodelor matematice, geometrice, statistice, ale fizicii si a teoriei invatarii automate** (machine learning)

Se bazeaza pe:

- Cunoasterea profunda a modelului camerei si al procesului de formare al imaginii pentru a obtine inferente simple de la valorile pixelilor individuali pana la combinarea informatiei de la imagini multiple pentru a obtine un tot unitar coerent
- Impunerea anumitor **ordonari asupra unor grupe de pixeli** pentru ai **separa sau grupa intre ei** sau pentru a **infera informatia de forma si a recunoaste obiecte** pe baza trasaturilor geometrice.

Alte denumiri

- analiza de imagini (image analysis)
- analiza scenei (scene analysis)
- interpretarea imaginilor (image understanding)



Viziunea artificiala

Disipline conexe

- Inteligenta artificiala (artificial intelligence)
- Robotica (robotics)
- Procesarea semnalelor (signal processing)
- Recunoasterea de forme (pattern recognition)
- Teoria controlului (control theory)
- Psihologia (psychology)
- Neurostiintele (neuroscience)

Subdomenii:

- Procesarea imaginilor
- Recunoasterea formelor
- Fotogrametria (discipline care se ocupa cu inferarea informatiei metrice /masuratori: calibrarea camerelor, stereoviziunea, reconstrucția 3D detectia de obiecte)
- Analiza statica/dinamica a imaginilor, trasaturilor, obiectelor



Procesarea imaginilor

Procesarea imaginilor (Image Processing)

- Se ocupa cu *studiul proprietatilor imaginilor* si cu *transformarea acestora* (a imaginilor)
- Majoritatea algoritmilor de viziune artificiala necesita folosirea unor algoritmi de procesare a imaginilor

Example de metode:

- imbunatatirea calitatii imaginilor (*image enhancement*) – prin transformarea imaginilor: punerea in evidenta a detaliilor ascune/obscure, a trasaturilor de interes
- *compresia* (reprezentare compacta a imginilor/secventelor pentru transmisie / stocare)
- *restaurarea* (eliminarea elementelor de degradare cunoscute / modelabile)
- *extragerea de trasaturi* (localizarea anumitor sabloane – ex: muchii, colturi, structuri complexe - obiecte)



Viziunea artificiala

Domenii de cercetare:

- Detectia de trasaturi (Feature Detection)
- Representarea conturelor (Contour Representation)
- Analiza imaginilor de profunzime (Range image analysis)
- Modelarea si reprezentarea formelor (Shape modeling and representation)
- Stereo viziunea (Stereo vision)
- Viziunea color (Color vision)
- Vizunea activa (Active/Purposive vision)
- Invarianti (Invariants)
- Detectia obiectelor (Object detection)
- Analiza miscarii (Motion analysis)
- Urmarirea miscarii obiectelor (Tracking)
- Recunoasterea obiectelor 3D (3D object recognition)
- Adnotarea semantica a imaginilor/video (Image annotation)
- Aritectura sistemelor de viziune (Vision architectures)



Viziunea artificiala

Domenii de aplicare

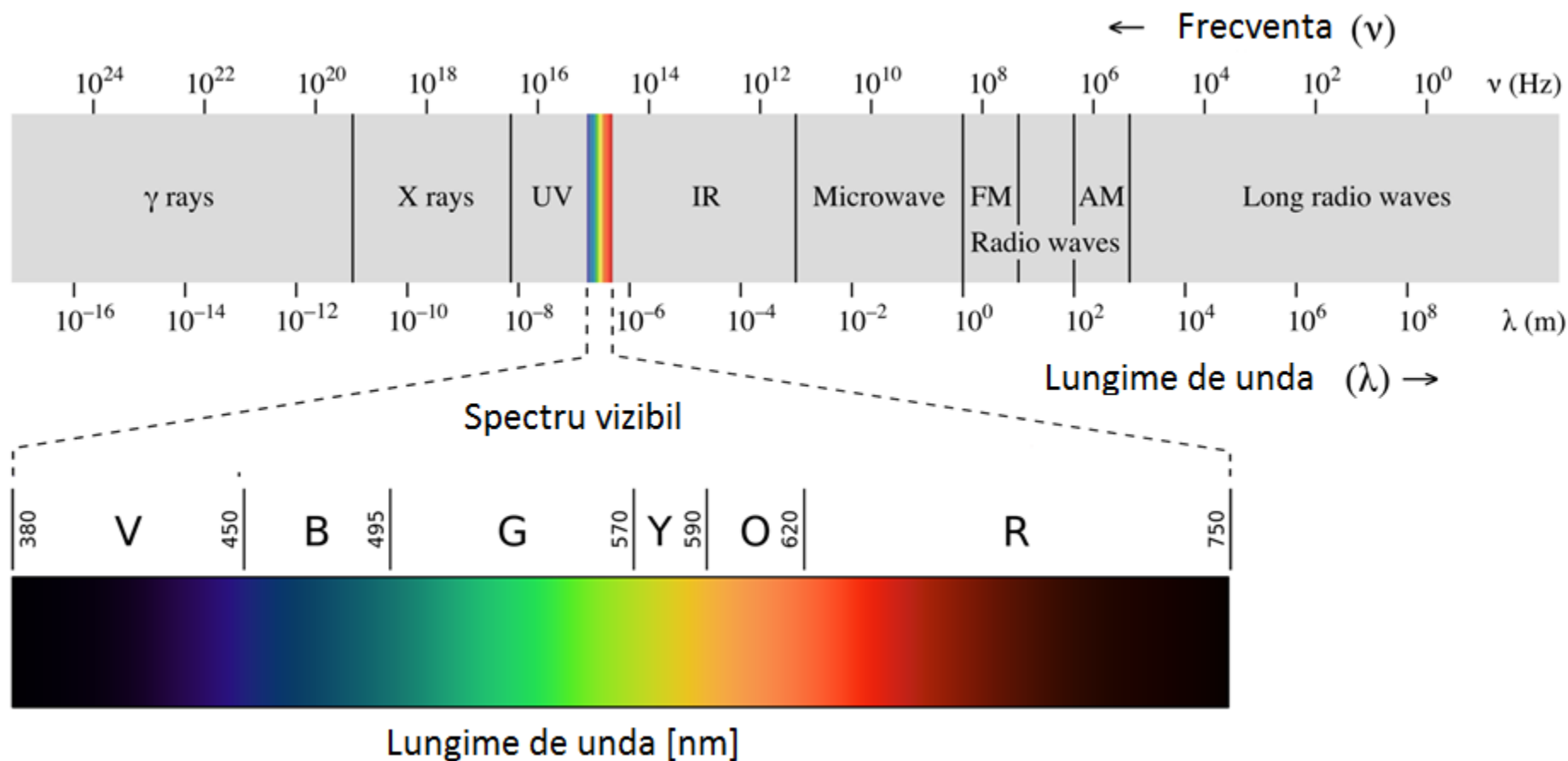
- Inspectie industrială / controlul calitatii (Industrial inspection/quality control)
- Inginerie inversa (reverse engineering)
- Supraveghere si securitate (Surveillance and security)
- Aplicatii biometrice:recunoastera fetei (Face recognition), amprentelor ..)
- Interactiune om-calculator: recunoasterea gesturilor (Gesture recognition)
- Monitorizarea traficului (Road monitoring)
- Aplicatii spatiale (Space applications)
- Analiza imaginilor medicale (Medical image analysis)
- Realitate virtuala / augmentata, teleprezenta si telerobotica (Virtual / augmented reality, telepresence, and telerobotics)
- Vehicule autonome (Autonomous vehicles)
- Cartografiere automata, achizitie automata de modele (Automated map making, model acquisition)



Viziunea artificiala

Date de intrare

- Imagini captate cu dispozitive de achizitie adaptate pentru intregul spectru frecventa al *undelor electromagnetice*



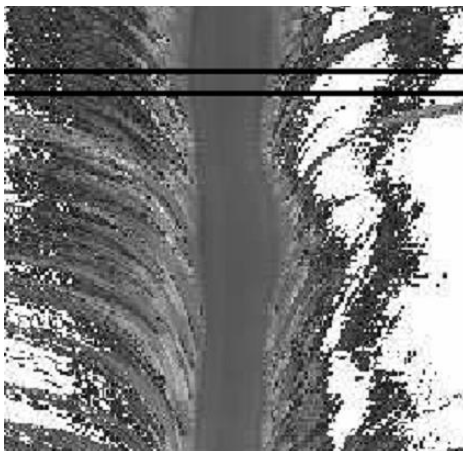


Viziunea artificială

Date de intrare



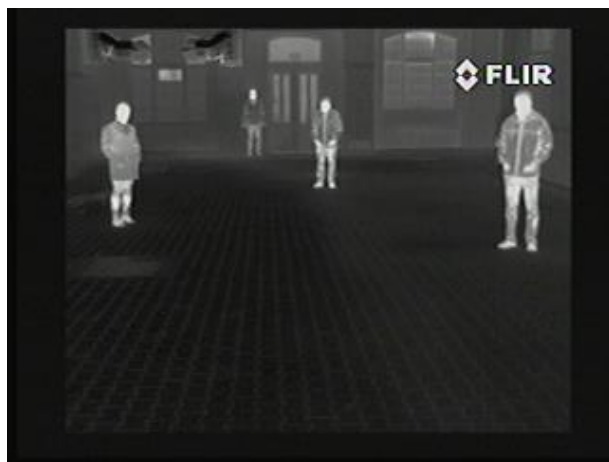
Spectrul vizibil (380 – 750 nm)



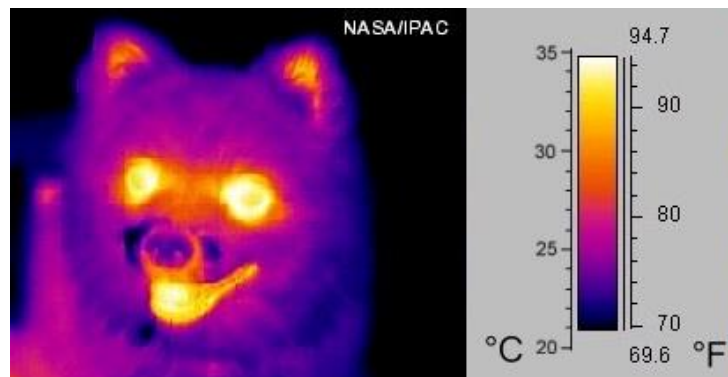
Reflectivitatea LiDAR
(NIR : 0.75 .. 1.4μm)



Reflectivitatea RADAR
24GHz (SRR) / 77 (LRR)



LWIR / Thermal (8-15μm)



LWIR / Thermal (heatmap)



Viziunea artificiala

Date de intrare

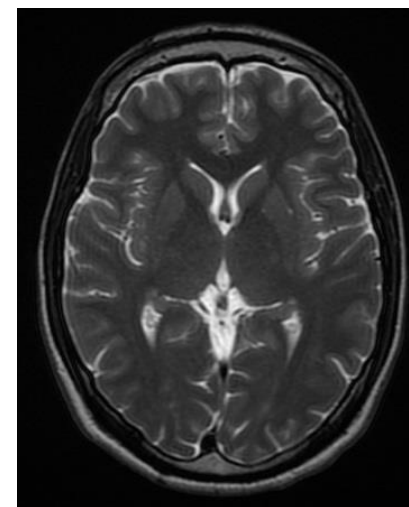
Alte surse de imagini: *unde acustice (US / ecografii), RMN, CT*



US 7–18 MHz: muschi, tiroida, ..
US 1–6 MHz: ficat, rinichi, ...



X-ray (radiografie)



MRI (RMN)



X-ray (CT slice)

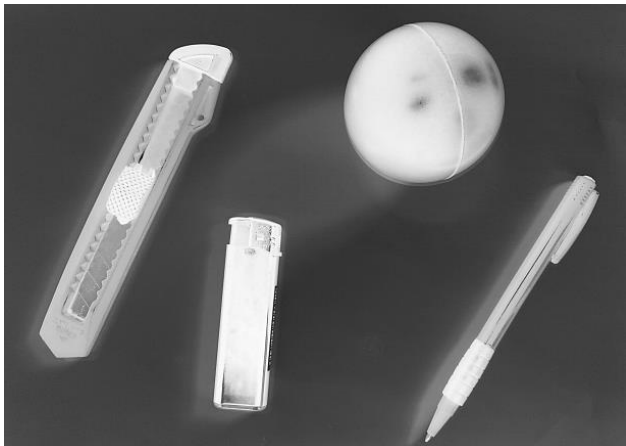
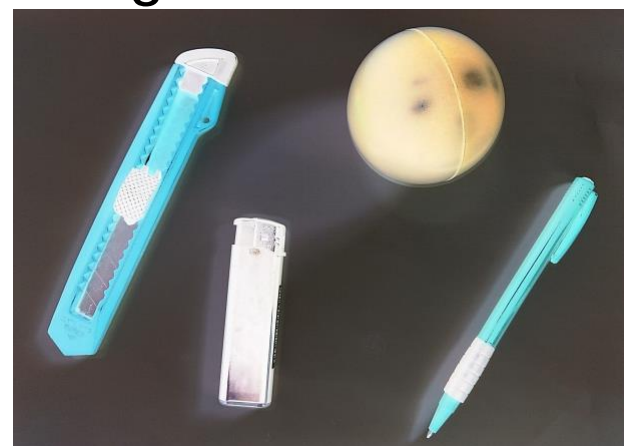


Exemple de procesări simple

Conversii Color - Nivele de gri – Imagine binară



Negativ





Exemple de procesări simple

Modificare strălucire și contrast



Luminozitate ++



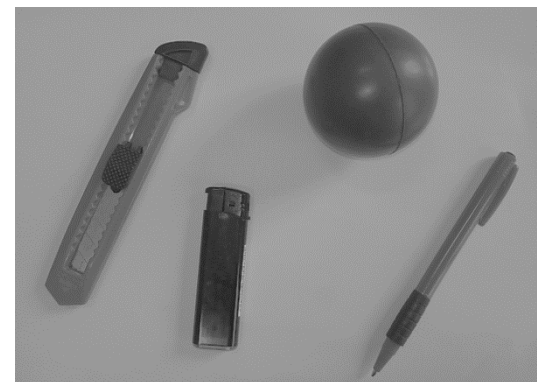
Imaginea sursa



Contrast ++



Luminozitate --

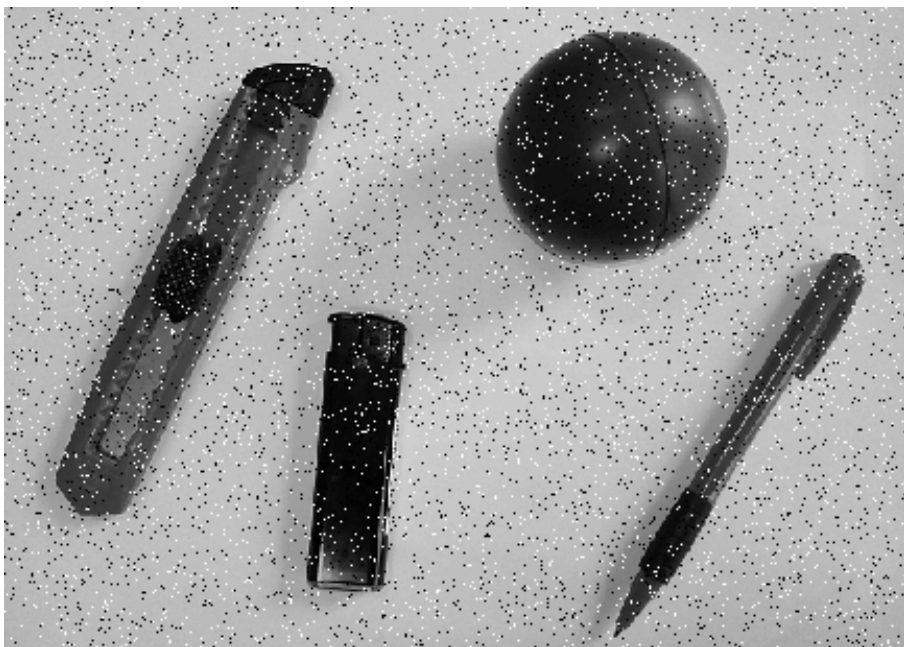


Contrast --



Exemple de procesări simple

Eliminarea zgomotului



Imagine corupta de zgomot
de tip sare și piper

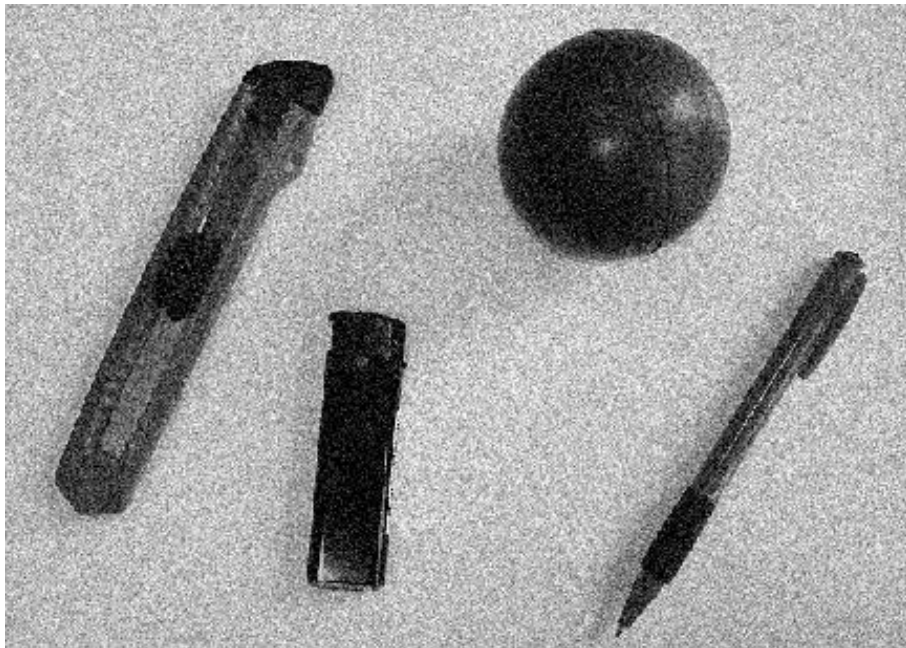


Imagine filtrata: filtru median

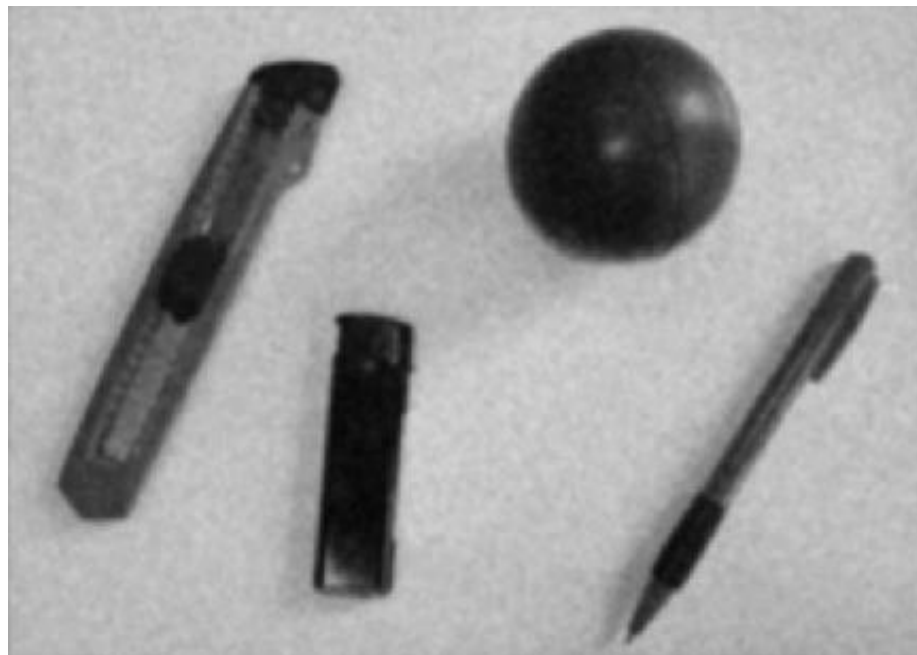


Exemple de procesări simple

Eliminarea zgomotului



Imagine corupta de Zgomot Gaussian

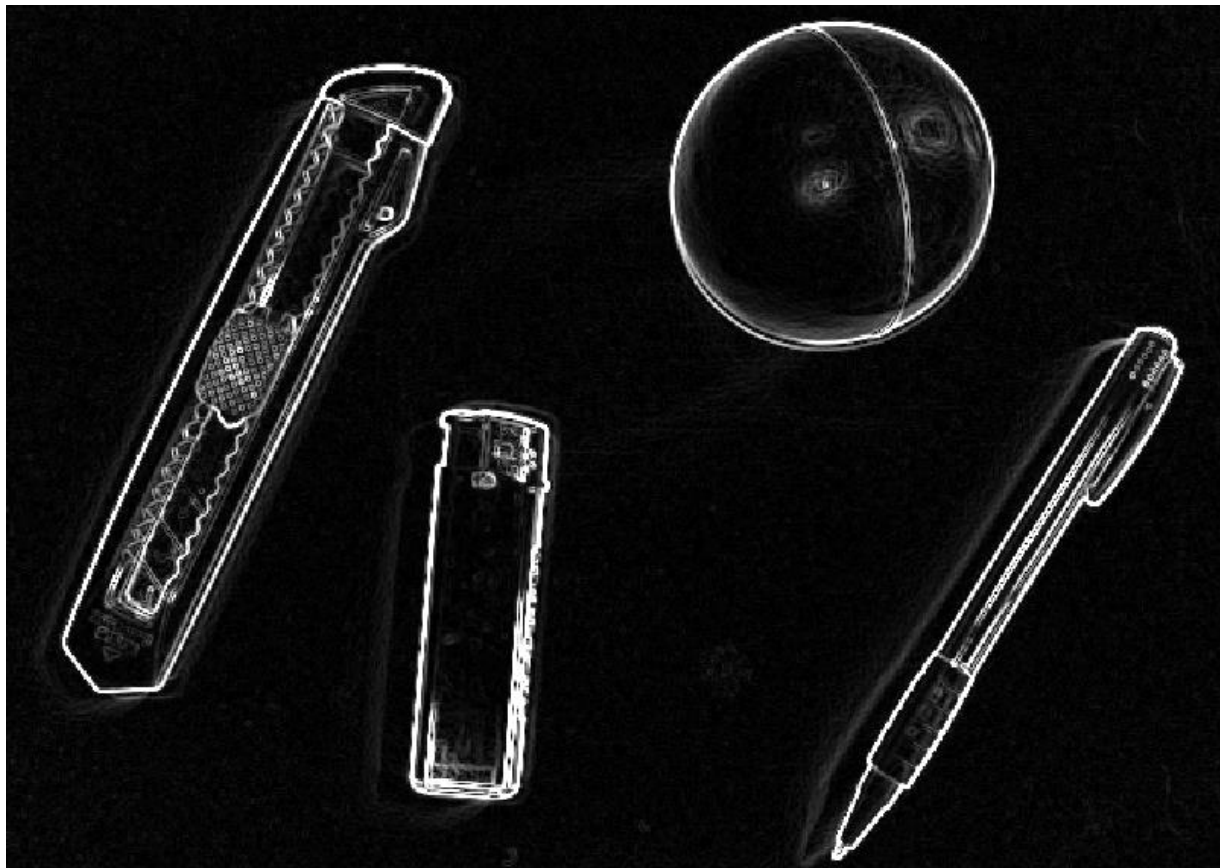


Imagine filtrata: filtru Gaussian



Exemple de procesări simple

Calculul gradientului – măsura variațiilor din imagine

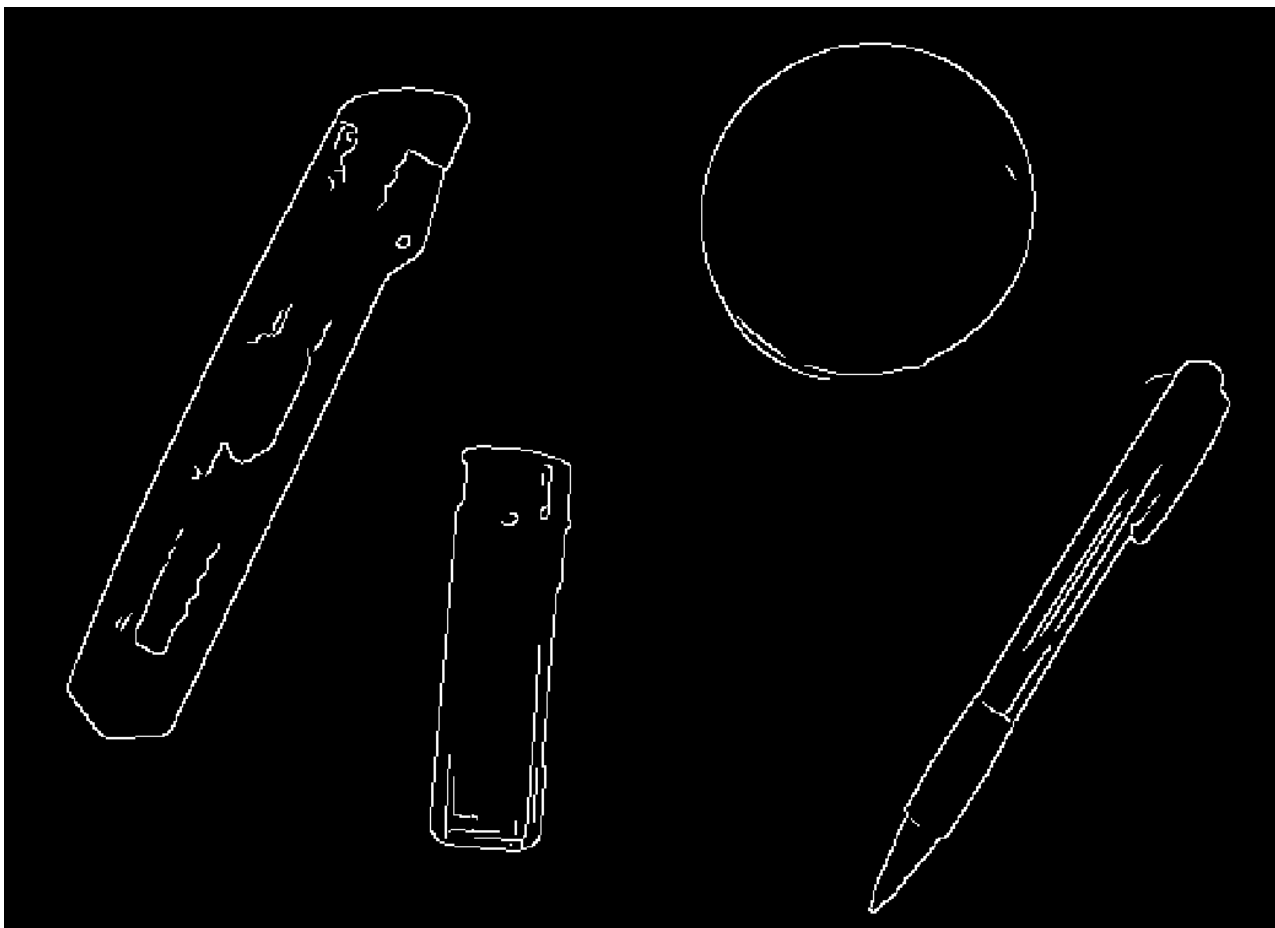


Modulul (magnitudinea gradientului)



Exemple de procesări simple

Determinarea punctelor de muchie



Rezultatul aplicarii alg. Canny (cubtiere + binarizare cu histereza)

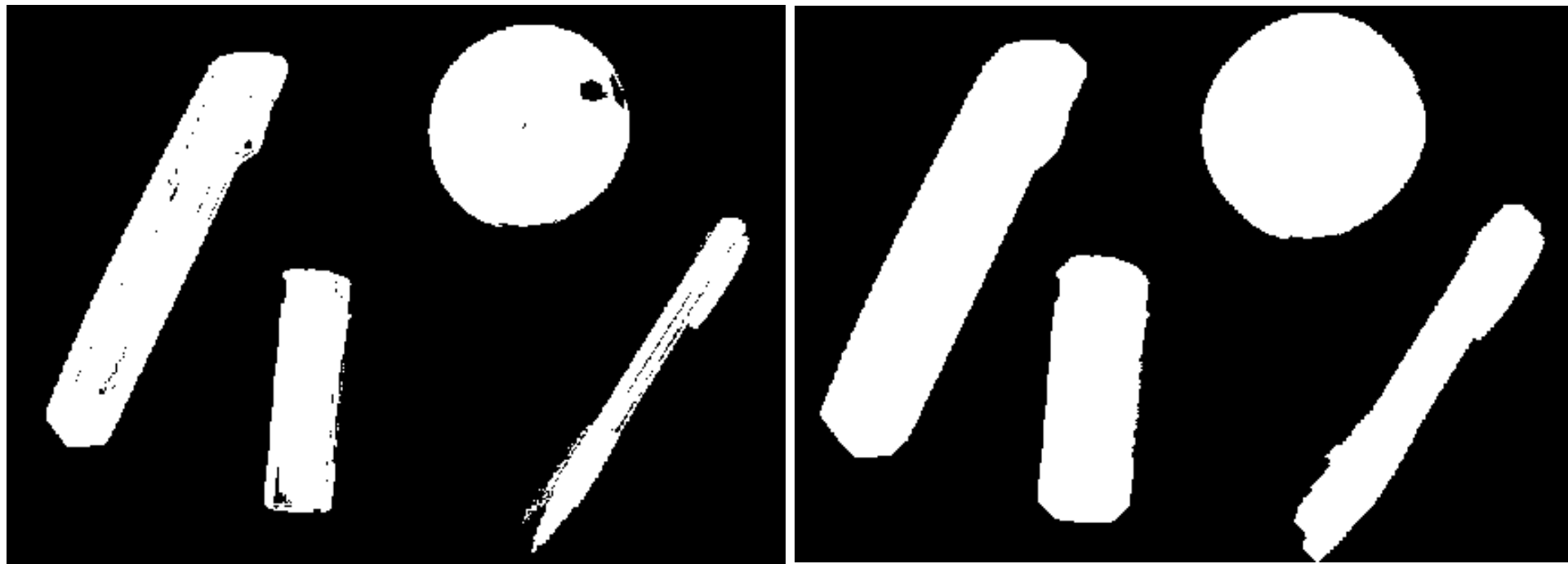
Technical University of Cluj Napoca

Computer Science Department



Exemple de procesări simple

Operații pe imagini binare - dilatare





Exemple de procesări simple

Operații pe imagini binare - eroziune





Exemple de procesări simple

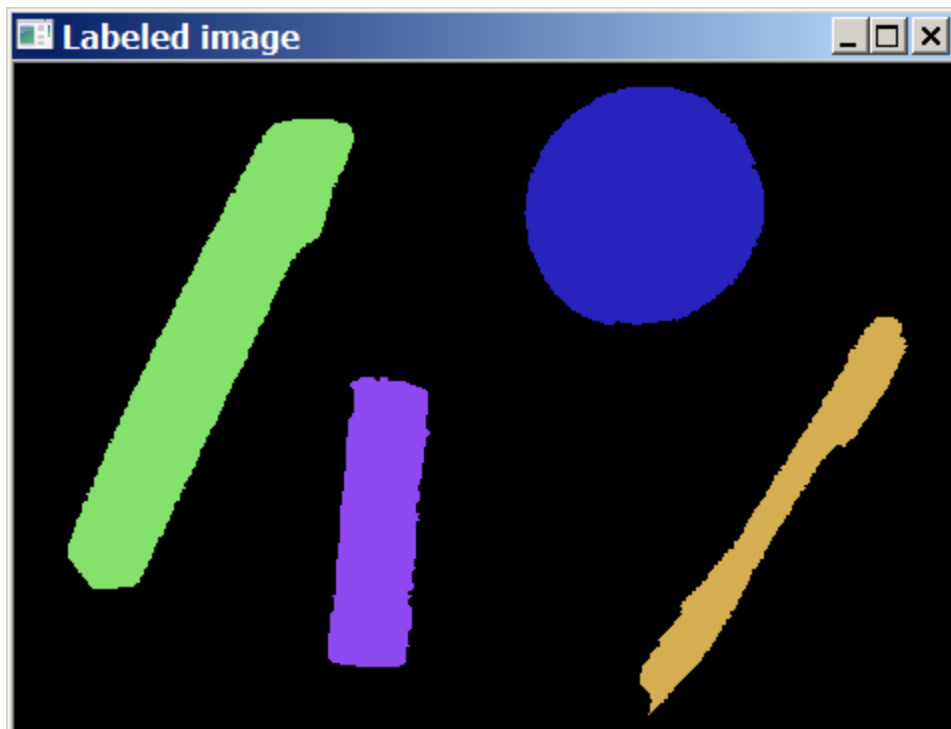
Dilatare + eroziune = închidere, umplerea golurilor cu păstrarea conturului exterior neschimbat





Exemple de procesări simple

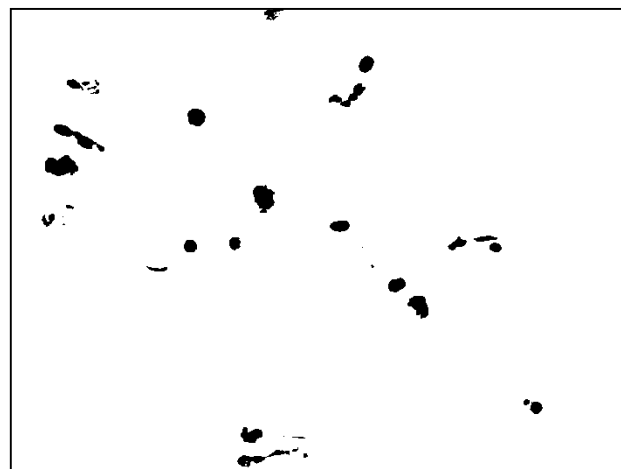
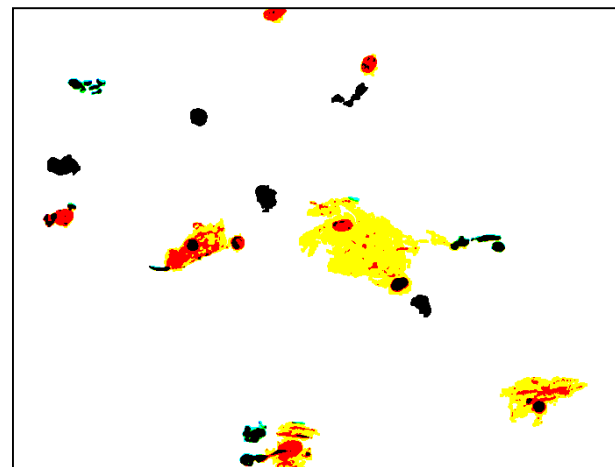
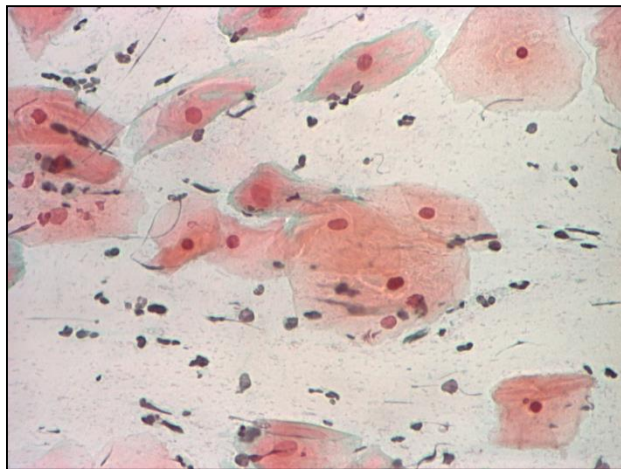
Identificarea obiectelor individuale din imagini binare -
etichetarea





Exemple de procesari complexe

Segmentare si analiza de imagini medicale



Features

The features extracted from the abnormal nuclei found in the image are:

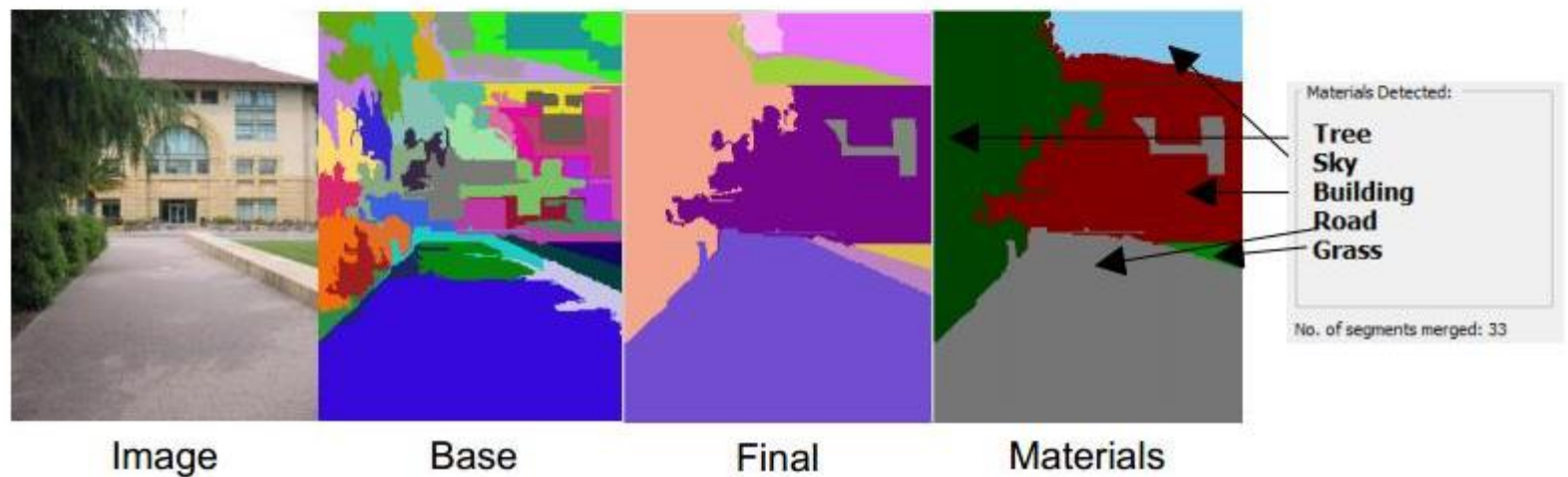
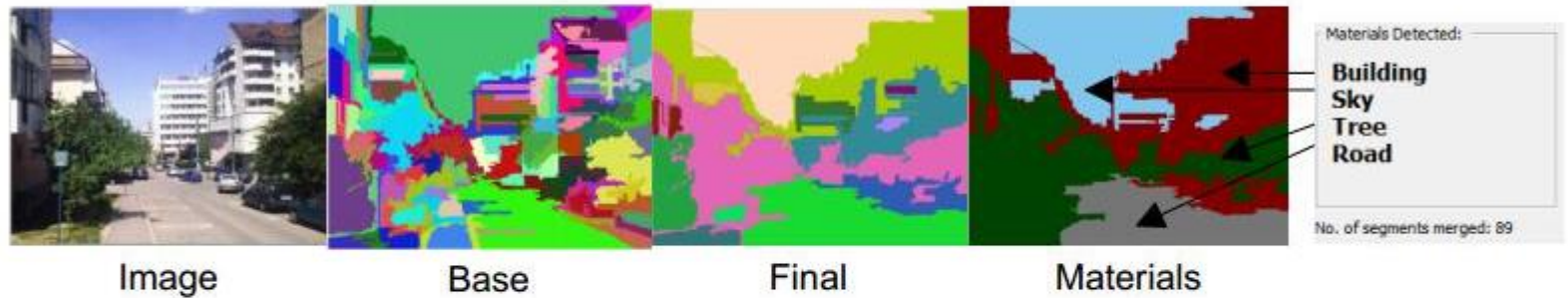
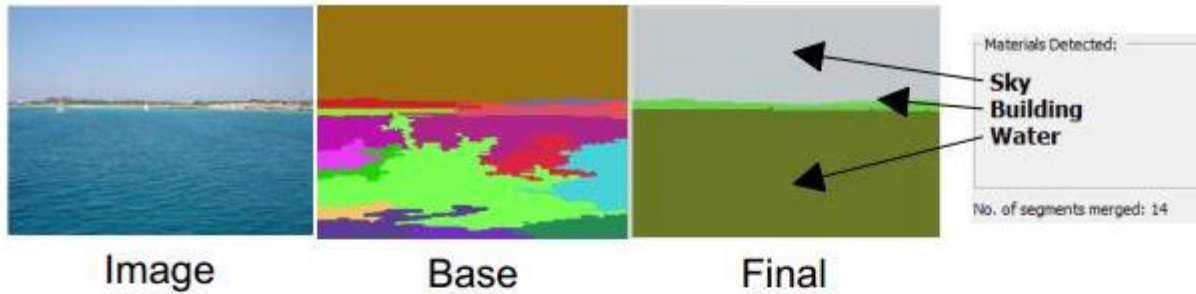
Nucleus	Entropy	Homogeneity	Coarseness	Busyness	Fractal dimension	No. clusters	No. colors	
0	2.53647	0.450885	0.0131579	0.661315	1.96163	20	2441	
1	2.57566	0.404834	0.0227273	0.611727	4.71926	12	1027	
2	2.78164	0.438416	0.013245	0.660755	2.14959	19	2436	
3	2.47713	0.454494	0.0168067	0.639943	4.65406	16	1230	
4	2.51205	0.435782	0.016	0.644344	2.2737	16	1548	
5	2.48444	0.442439	0.018018	0.633624	4.3415	12	1310	

The execution times for the operation mentioned below are:

Operation	Time elapsed
KSW on RGB	00:00:735
KSW and Minimum Error	00:04:422
MultiSpace thresholding	00:02:156
HMT	00:00:390
Texture-related features	00:13:750



Segmentare imagini

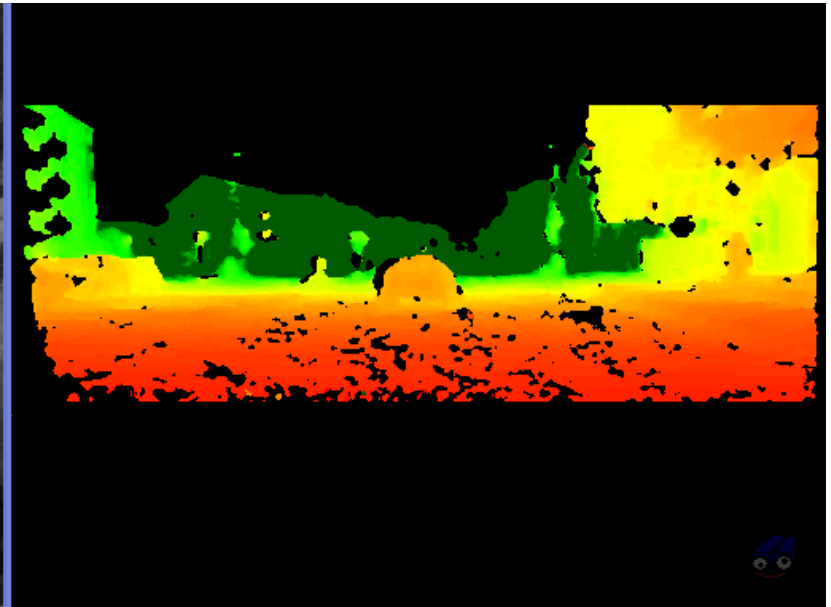




Stereoviziune densa



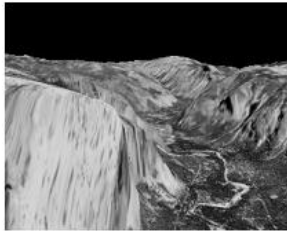
Imagine camera
stanga (grayscale)



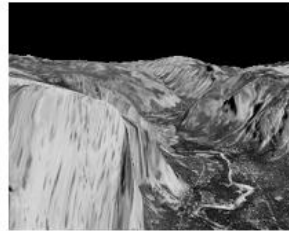
Harta de adancime



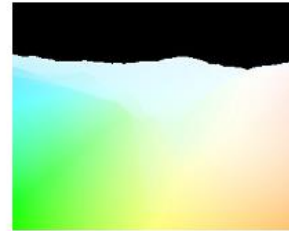
Estimarea fluxului optic



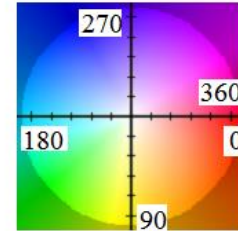
Yosemite frame 0



Yosemite frame 1



Yosemite GT flow



flow color coding

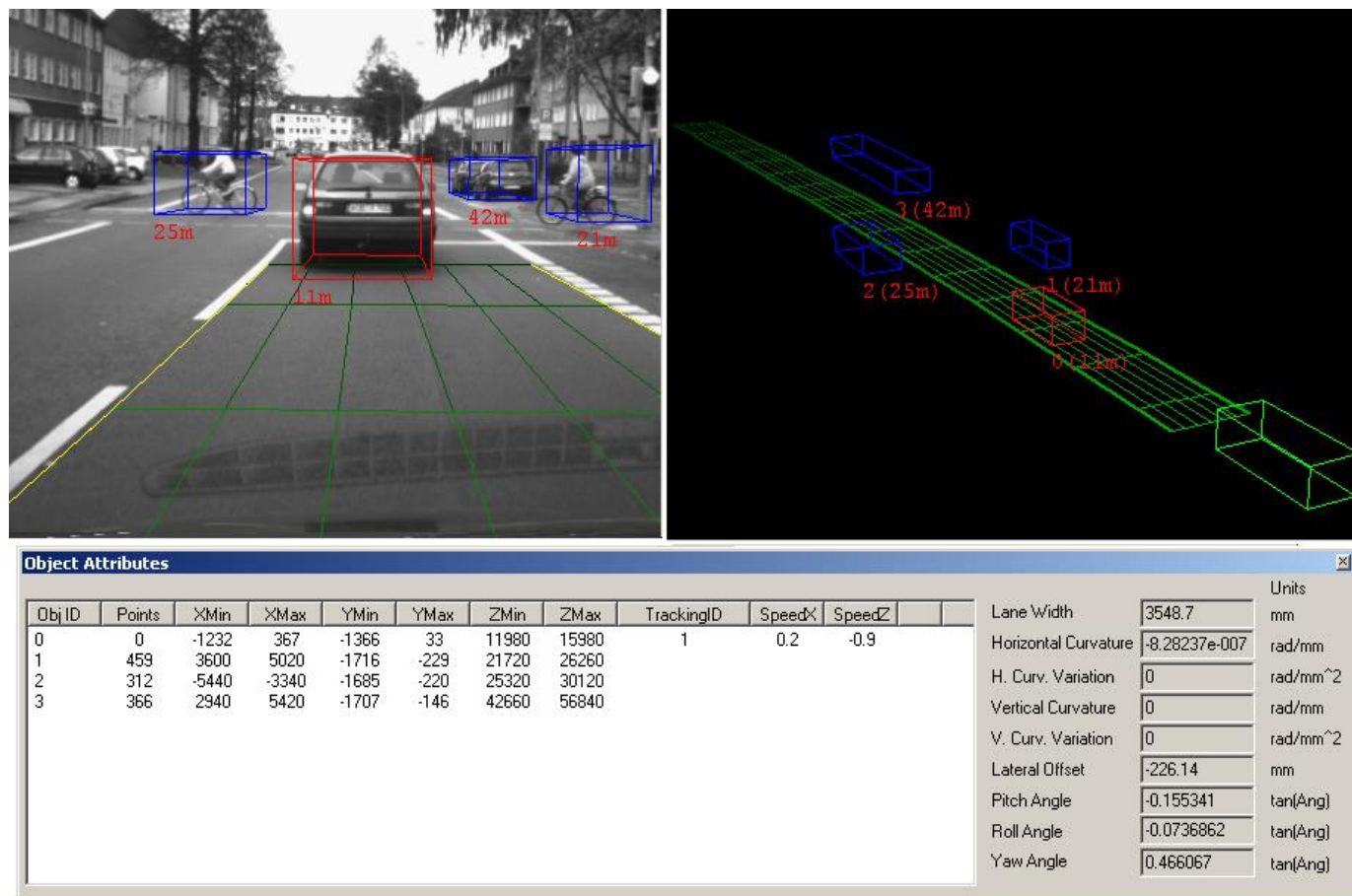
Exemplificare a rezultatului codificarii fluxului de miscare in conformitate cu conventia de culoare Middleburry [4]:
Hue – directia / Saturatia – amplitudinea (similar cu codificarea culorilor in modelul HIS)





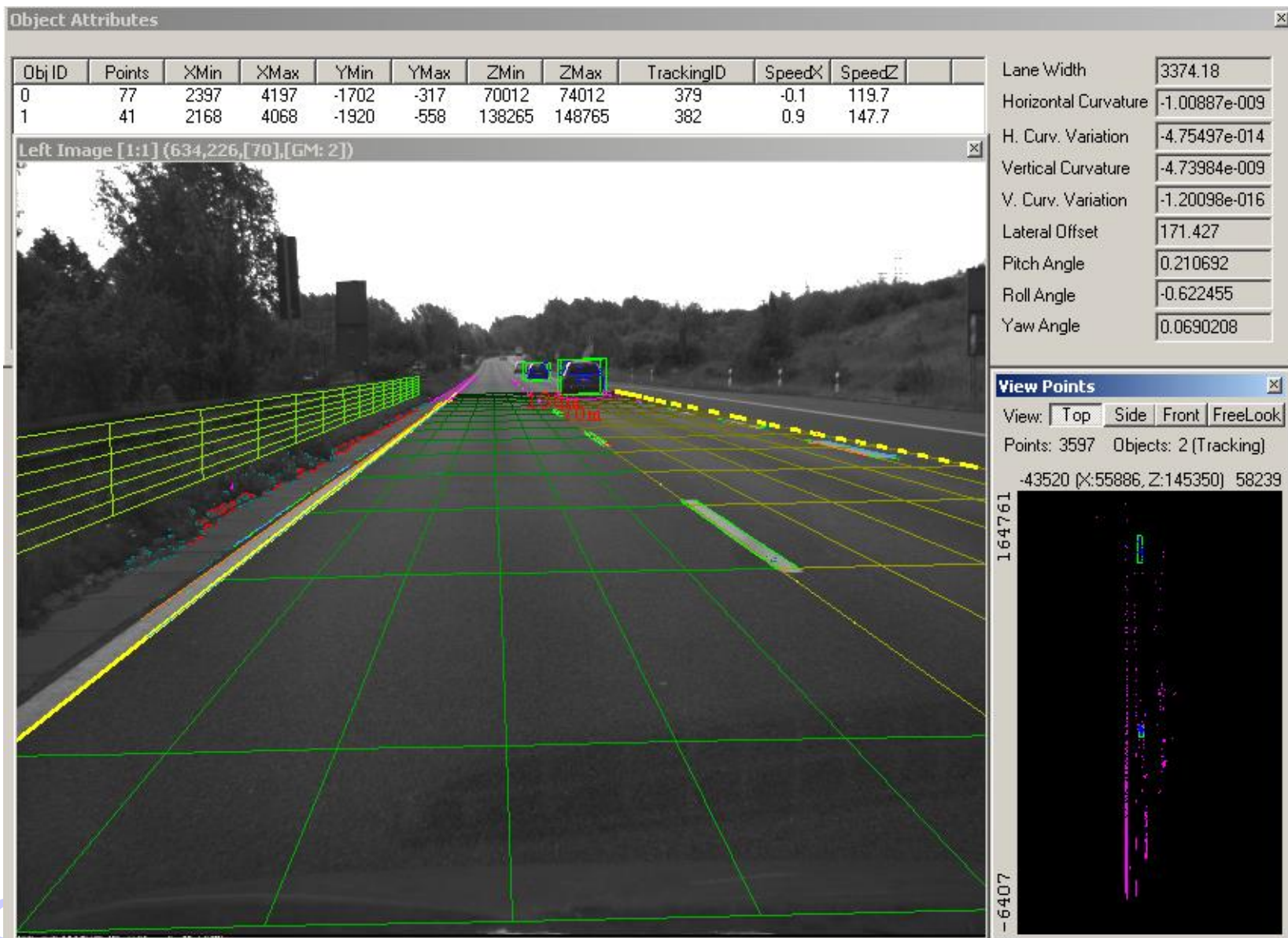
Exemple de aplicatii

Detectie obiecte si drum in scenarii de trafic





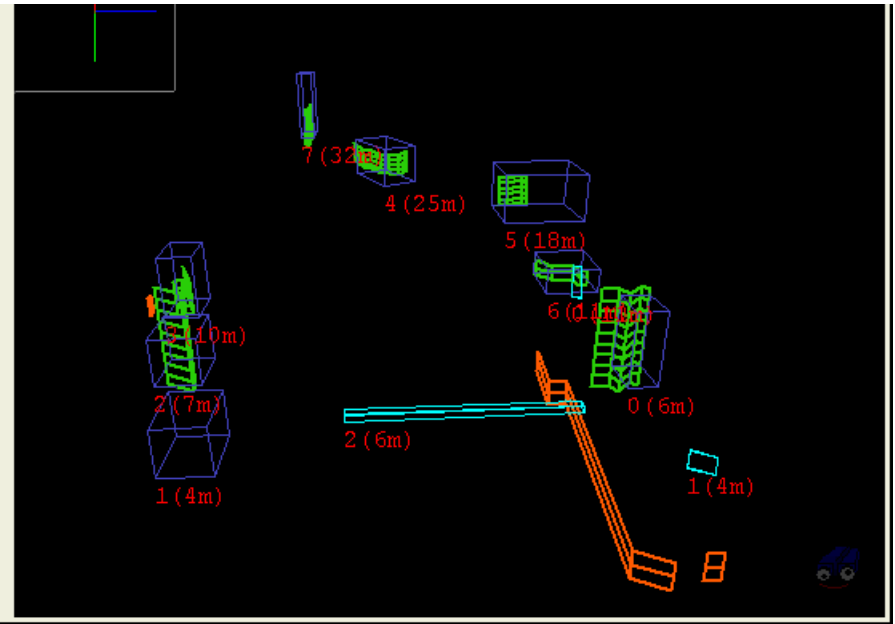
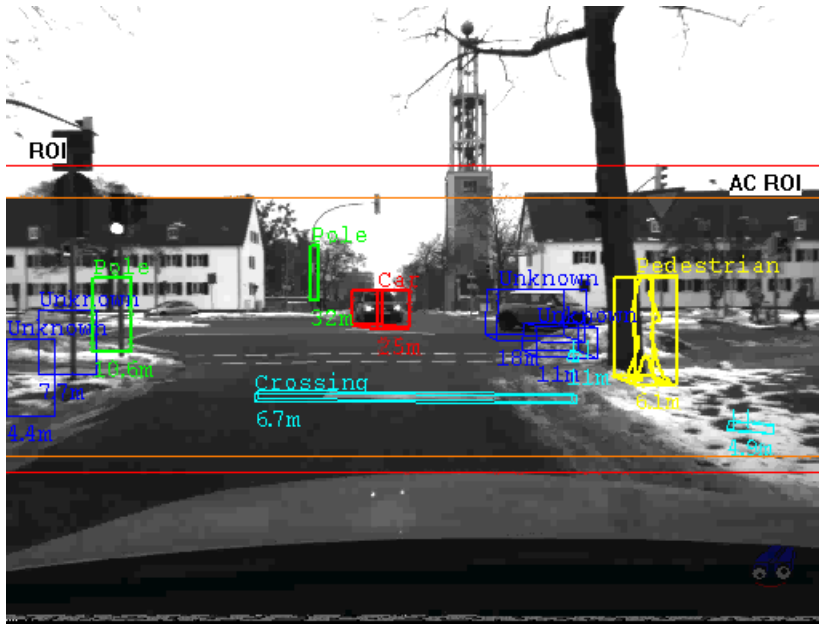
Exemple de aplicatii





Exemple de aplicatii

Detectie si clasificare obiecte, marcaje de pe drum etc. in scenarii de trafic





Proiecte de cercetare in domeniu

Image Processing and Pattern Recognition Group

<http://www.cv.utcluj.ro/research.72.html>

Driving Assistance Systems

SCABOR, DESBOR, DESPED, PERSENS: <http://www.cv.utcluj.ro/scabor.html>

IntrSAFE FP7 project: <http://www.cv.utcluj.ro/intersafe-2.html>

Optical Flow: <http://www.cv.utcluj.ro/optical-flow.html>

Medical Imaging

<http://www.cv.utcluj.ro/tdii.html>

<http://www.cv.utcluj.ro/focal-liver-diseases.html>

<http://www.cv.utcluj.ro/adkp-detection.html>

Etc.



Procesarea imaginilor

Discipline conexe in ani superiori

Licenta

- Sisteme de recunoastere a formelor (an 4 CA)
- Interactiune om-calculator (an 4 TI)

Master

- Viziune Artificiala (an 1 – toate specializarile)
- Viziune Artificiala pentru roboti mobili (an 2 – IVA)
- Interfete om-calculator (an 2 – IC)